Transformer equivalent circuit

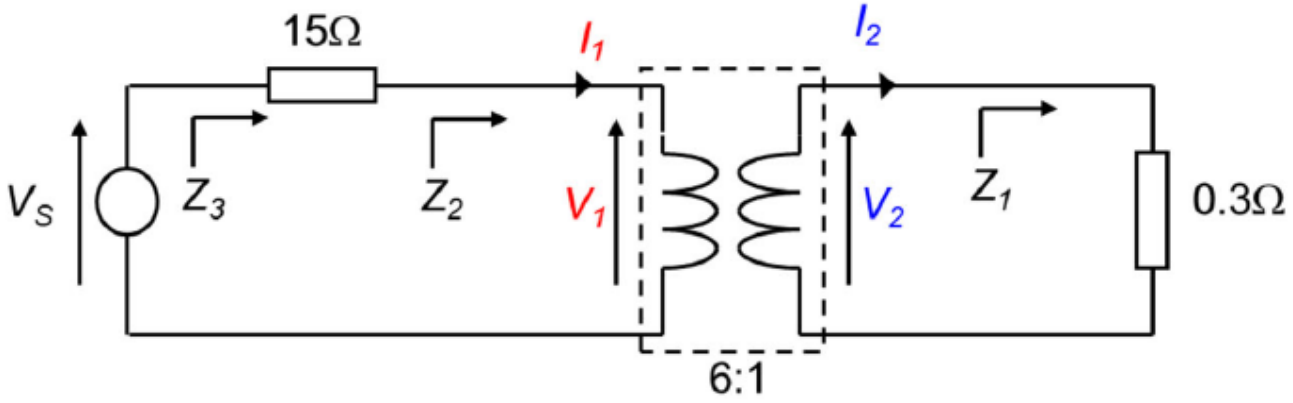
Practice questions · Day 04 · Tue 27 Oct · pack M4

5. Practice

From your course files

Jun24 Q2b (past exam, 5 marks). *Ideal Transformers.* In the diagram on the right the transformer is ideal and has a turns ratio of 6:1.

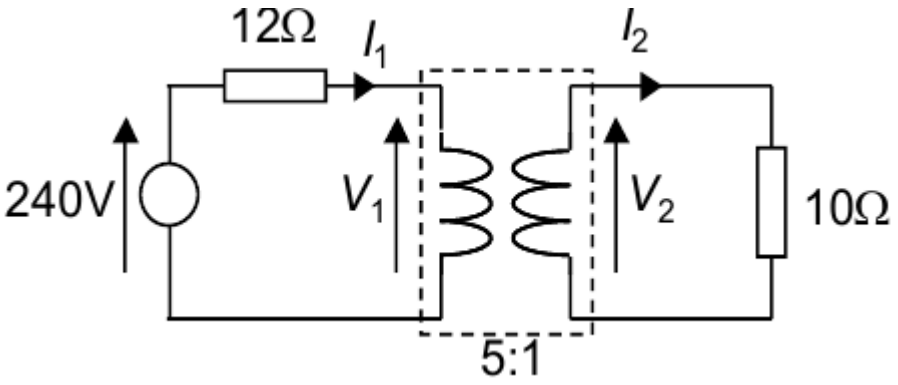
i. Find the values of the impedances Z_1, Z_2 and Z_3 shown in the figure. [2 marks]
ii. Assuming the AC supply voltage V_S is 12 V, find the magnitudes of the primary current I_1 and the actual (not referred) secondary current I_2. Find also the magnitudes of the primary voltage V_1 and the actual secondary voltage V_2. [3 marks]



Jun24 Q2b figure: V_S, 15 Ω series resistor, ideal 6:1 transformer and 0.3 Ω load, with Z_3, Z_2 and Z_1 marked (from the exam paper)

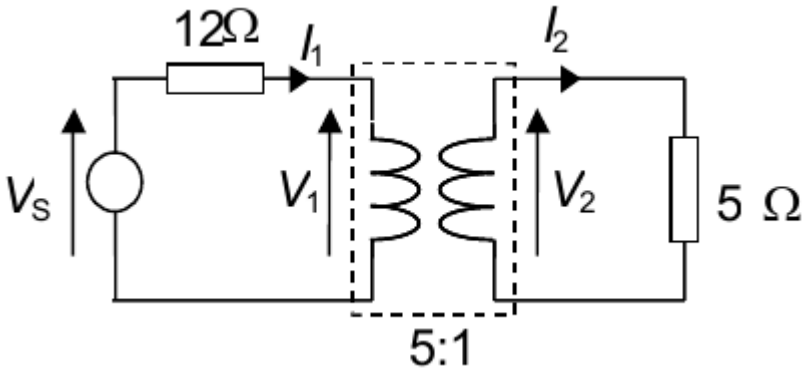
PQ 3.20 (practice questions). Consider the ideal transformer circuit shown below.

a. Find the load resistance referred to the primary.
b. Calculate the primary current I_1. [hint: it may be helpful to draw the equivalent circuit using the load resistance referred to the primary.]
c. Find the primary voltage V_1 [hint: Ohm's law]
d. Find the secondary current and voltage.
e. Calculate the output power delivered to the load
f. Calculate the power loss in the 12 Ω resistor.
g. Calculate the efficiency of power transfer from the voltage source to the load.



PQ 3.20 figure: 240 V source, 12 Ω series resistor, ideal 5:1 transformer and 10 Ω load (from the practice questions)

PQ 3.21 (practice questions). Consider the ideal transformer circuit shown below. The input voltage is 240 V. Calculate the secondary voltage.



PQ 3.21 figure: source V_S, 12 Ω series resistor, ideal 5:1 transformer and 5 Ω load (from the practice questions)

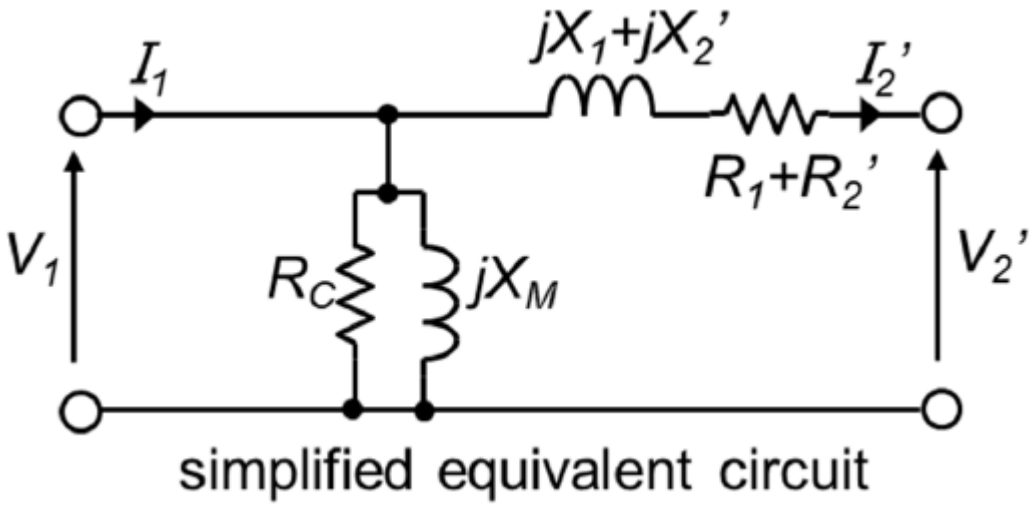
Jun23 Q2b (past exam, 5 marks). *Transformer Parameters.* The following tests are performed on a single-phase 240 V:32 V, 2400 VA transformer.

- Short-circuit test. Primary measurements are 27 V, 9.7 A, and 55 W.
- Open-circuit test. Primary measurements are 240 V, 1.3 A, and 45 W.

Find all the simplified equivalent circuit parameters.

Jun23 Q2c (past exam, 6 marks). *Transformer Analysis.* Consider the transformer simplified equivalent circuit shown on the right. The values of the primary voltage $\mathbf{V_1}$ and current $\mathbf{I_1}$ phasors are known and it is desired to find the value of the secondary voltage $\mathbf{V_2'}$ phasor.

i. Using Kirchhoff's voltage law (KVL), write an expression for $\mathbf{V_2'}$ in terms of $\mathbf{V_1}$, $\mathbf{I_2'}$ and the relevant equivalent circuit parameters shown. [2 marks]
ii. Assuming the load is purely resistive with a value of R_L', use the voltage divider rule to write an expression for $\mathbf{V_2'}$ in terms of $\mathbf{V_1}$ and the relevant equivalent circuit parameters shown. [2 marks]
iii. For a purely resistive load, would you expect that $\mathbf{V_2'}$ will be leading, in phase or lagging with respect to $\mathbf{V_1}$? Carefully explain why. [2 marks]



Jun23 Q2c figure: simplified equivalent circuit with R_C and jX_M across V_1, then jX_1+jX_2′ and R_1+R_2′ in series to V_2′ (from the exam paper)

PQ 3.27 (practice questions). A single-phase 1000 VA, 240 V/48 V transformer has the following parameters: primary resistance 0.6 ohm, secondary resistance 0.02 ohms, primary leakage reactance 2 ohms, secondary leakage reactance 0.06 ohms, magnetising reactance 550 ohms, core-loss resistance 3200 ohms. Note the secondary parameters are NOT referred to the primary. It is connected to a 240 V, 50 Hz supply for parts (a) to (c).

a. Draw a simplified equivalent circuit for the transformer referred to the primary and calculate all the circuit parameters.
b. Find the no-load (that is, with the secondary winding open-circuited) input current magnitude and input power at rated voltage.
c. Find the input current magnitude and input power at rated input voltage with the secondary winding short-circuited. You can neglect the core-loss resistance and the magnetising reactance in this part and for part (d).
d. Find the rated input current. Then find the required supply voltage magnitude to produce rated input current with the secondary winding short-circuited.

PQ 3.28 (practice questions). Using the input voltage and current magnitudes and input power results from parts (b) and (c) of the above question, calculate the reactances of the simplified equivalent circuit. Verify the calculated parameters are identical to the actual parameters.

PQ 3.29 (practice questions, do (a)–(c)). The following tests are performed on a single-phase 240 V:24 V, 2400 VA transformer. Short-circuit test. Secondary winding is short-circuited. Primary measurements are 27 V, 9.5 A, and 55 W. Open-circuit test. Primary winding voltage is 240 V, current is 1.2 A, power is 45 W.

a. Find the simplified equivalent circuit parameters.
b. The transformer is delivering rated output at a power-factor of 0.87 lagging at rated output voltage. Calculate the secondary voltage and current referred to the primary.
c. Find the required primary voltage corresponding to the case in part (b).
d. Find the output power, copper loss, iron loss and efficiency of the transformer corresponding to the case in part (b).

2024 test Q6 (4 marks). *Transformer Parameters.* Consider a star-connected, three-phase transformer. The following test results were obtained:

- open-circuit test: 5.2 kV, 45 A, 49 kW
- short-circuit test: 710 V, 390 A, 140 kW

Find the winding resistance and the leakage reactance for the simplified transformer equivalent circuit. It is not necessary to find the other equivalent circuit parameters.

2024 test Q7 (4 marks). *Transformer Analysis.* A three-phase, 14 kV:135 kV 250 MVA transformer is supplying a load. The load has an apparent power of 150 MVA and is generating 35 MVAr of reactive power. The output voltage is equal to its rated value. Find the secondary phase voltage $\mathbf{V}_2'$ and phase current $\mathbf{I}_2'$ phasors referred to the primary. Assume the phase angle of $\mathbf{V}_2'$ is zero. Hint: remember the power triangle!

2024 test Q8 (4 marks). *Short-Answer Question.* Answer the following question using words only (that is, without using sketches or equations).

Consider an ideal transformer driving a resistive load of resistance R. Carefully explain what would be the effect on the output voltage and current if the load resistance was doubled. Carefully explain why and how this would be different for a real transformer.

2023 quiz Q6 (9 marks). *Transformer Analysis.* A three-phase 50 kVA, 3.3 kV/415 V transformer has the following simplified single-phase equivalent circuit parameters referred to the primary: total leakage reactance 35 ohms, total winding resistance 4 ohms, magnetising reactance 2,300 ohms, core loss resistance 18,000 ohms. The transformer is delivering 27 kW at a power-factor of 0.78 lagging at rated output voltage.

a. Calculate the magnitude and phase of the secondary phase voltage $\mathbf{V}_2'$ and current $\mathbf{I}_2'$ referred to the primary. [4 marks]
b. Under a different load condition, it is known that the secondary phase voltage $\mathbf{V}_2' = 2100\angle 0^\circ$ V and that the secondary phase current $\mathbf{I}_2' = 7.7\angle -25^\circ$ A (both referred to the primary). Draw a simplified equivalent circuit for the transformer referred to the primary and hence calculate the required primary (line) voltage magnitude. [5 marks]

PQ 3.30 (practice questions, overflow: do (a), (b)). A three-phase 3000 VA, 415 V/48 V transformer has the following simplified single-phase equivalent circuit parameters referred to the primary: total leakage reactance X 8.5 ohms, total winding resistance R 0.95 ohms, magnetising reactance X_M 510 ohms and core loss resistance R_C 3300 ohms. The transformer is delivering rated output at unity power-factor at rated output voltage.

a. Calculate the secondary phase voltage and current referred to the primary.
b. Calculate the required supply voltage magnitude.
c. Calculate both the total copper winding loss and total core loss.
d. Hence calculate the transformer efficiency.

PQ 3.31 (practice questions, overflow: do (a), (b)). The following tests are performed on a three-phase 15 kV:135 kV, 100 MVA star-connected transformer. Short-circuit test with secondary winding short-circuited. Primary measurements are 95 V, 500 A, and 12 kW. In the open-circuit test the primary winding voltage is 415 V, current is 15 A, power is 800 W.

a. Find the simplified equivalent circuit parameters referred to the primary winding.
b. Assuming the transformer is delivering 50 MW at a power-factor of 0.8 lagging at rated output voltage, calculate the secondary phase voltage and current referred to the primary and hence the required input voltage.
c. The transformer efficiency corresponding to the operating case described in part (b).

New questions

New Question 1 (easy) [6 marks]. A 240 V, 50 Hz source with an internal series resistance of 8 Ω feeds the primary of an ideal 8:1 transformer. The secondary supplies a load $Z_L = 0.4 + j0.3\,\Omega$.

a. Find the load impedance referred to the primary. [1]
b. Find I_1 and V_1 in polar form, taking the source at 0°. [3]
c. Find the actual secondary current and voltage, and the real power delivered to the load. [2]

New Question 2 OC test: meters on the 110 V winding, 415 V winding open

New Question 2 SC test: meters on the 415 V winding, 110 V winding shorted

New Question 2 (medium) [5 marks]. A single-phase 5 kVA, 415 V:110 V transformer is tested. The 415 V winding is the primary.

- Open-circuit test, done on the **110 V side** (415 V side open): 110 V, 1.8 A, 55 W.
- Short-circuit test, done on the **415 V side** (110 V side shorted): 22 V, 12.05 A, 120 W.

Find all the simplified equivalent circuit parameters referred to the 415 V primary. Draw the circuit.

New Question 3 (medium) [5 marks]. A 3-phase, star-connected 11 kV/415 V, 500 kVA transformer gives these primary-side test results:

- OC test: 11 kV, 1.2 A, 1.9 kW

- SC test: 480 V, 26.24 A, 5.2 kW

a. Find the per-phase simplified equivalent circuit parameters referred to the primary. [4]
b. Show that the SC test was carried out at rated current. [1]

Circuit for New Question 4: the New Question 2 transformer on 415∠0° V with the 2.42 Ω load referred to the primary

New Question 4 (hard, chained) [8 marks]. Use the transformer from New Question 2 (referred to the 415 V primary: $R = 0.8264\,\Omega$, $X = 1.628\,\Omega$, $R_C = 3131\,\Omega$, $X_M = 905.5\,\Omega$). Rated voltage, $415\angle 0^\circ$ V, is applied to the primary. A resistive load of **2.42 Ω** (actual value) is connected to the 110 V secondary.

a. Find the referred load resistance R_L'. [1]
b. Write the KVL expression for V_2' in terms of V_1, I_2' and the circuit parameters. [1]
c. Use the voltage divider rule to find V_2' in polar form, and the actual secondary voltage V_2. [3]
d. Find the primary current I_1 in polar form. [2]
e. A student claims that replacing the load with one of the same $|Z_L'|$ at 0.8 pf leading would make V_2' lead V_1. Check this numerically and comment. [1]

New Question 5 (hard, saturation) [5 marks]. A single-phase 240 V, 50 Hz transformer has two OC tests on its primary:

- at 120 V: 0.050 A, 2.2 W
- at 240 V: 0.42 A, 8.6 W

a. Find R_C and X_M at both voltages. [2]
b. A designer uses only the 120 V values to predict the no-load current and iron loss at 240 V. What do they predict, and how does that compare with the measurement? [2]
c. Explain the difference in one or two sentences, and say which values belong in the equivalent circuit for rated operation. [1]

6. Quick recall check

1. In the SC test, why can R_C and X_M be neglected?
2. A 10:1 ideal transformer has a 2 Ω load on its secondary. What impedance does the primary see?
3. Which formula gives X_M from the OC test? Which gives X from the SC test?
4. For a resistive load, does V_2' lead, lag or stay in phase with V_1? Why, in one line?
5. Star-connected 3-phase OC test: 6.6 kV, 3 A, 6 kW. What are V_1 and P_1 per phase?